

SMART HOSPITAL EMERGENCY ROOM ANALYTICS SYSTEM

Project Documentation

Abstract

The Smart Hospital Emergency Room Analytics System is designed to improve emergency healthcare management through data analytics and decision support techniques. Hospitals receive numerous patients daily, making it challenging to manage resources efficiently.

This project generates patient datasets, validates data, calculates patient priority scores, classifies patients based on urgency, predicts hospital resource requirements, detects peak hours, analyzes bed occupancy, and generates management reports.

The system provides insights that help hospital administrators optimize doctor allocation, bed utilization, and emergency response planning.

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1. Introduction

Emergency rooms play a critical role in healthcare services. Efficient patient management is essential for reducing waiting times and improving treatment quality.

This project utilizes Artificial Intelligence and Data Analytics concepts to analyze patient information and assist hospitals in making informed decisions regarding patient prioritization and resource allocation.

2. Objectives

The objectives of the project are:

- Generate hospital patient datasets.
 - Validate patient records.
 - Calculate patient priority scores.
 - Classify patients according to severity.
 - Analyze hospital workload.
 - Predict doctor and bed requirements.
 - Detect peak patient arrival periods.
 - Generate automated recommendations.
 - Create visual analytics dashboards.
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3. Problem Statement

Hospitals often face challenges such as:

- Long waiting times.
- Limited medical staff.
- Bed shortages.
- Increasing patient load.
- Emergency management difficulties.

The project aims to solve these challenges using data-driven analytics.

4. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum
- Storage: 1 GB free space

Software Requirements

- Python 3.x
 - Visual Studio Code
 - Streamlit
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - OpenPyXL
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5. Methodology

The project follows the following workflow:

1. Dataset Generation
 2. Data Validation
 3. Priority Score Calculation
 4. Patient Classification
 5. Queue Analysis
 6. Doctor Requirement Analysis
 7. Bed Occupancy Analysis
 8. Peak Hour Detection
 9. Statistical Analysis
 10. Resource Forecasting
 11. Report Generation
 12. Dashboard Creation
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6. Dataset Description

The dataset contains 1000 patient records.

Field	Description
Patient ID	Unique Patient Identifier
Age	Patient Age
Gender	Male/Female
Arrival Time	Arrival Hour
Heart Rate	BPM
Oxygen Level	Percentage
Temperature	Body Temperature
Blood Pressure	BP Reading
Severity Score	Severity Level
Waiting Time	Waiting Duration
Treatment Required Yes/No	

7. System Architecture

Dataset → Validation → Priority Calculation → Classification → Analytics → Forecasting → Reporting → Dashboard

8. Module Description

Module 1: Dataset Generation

Creates synthetic patient records using Python and NumPy.

Module 2: Data Validation

Checks:

- Invalid age
- Invalid oxygen level
- Invalid heart rate
- Missing values

Module 3: Priority Score Calculation

Priority Score Formula:

Priority Score = Severity Score $\times$ 5 + Oxygen Risk + Heart Rate Risk + Temperature Risk + Age Risk

Module 4: Patient Classification

Categories:

- Normal
- Moderate
- High Priority
- Critical

Module 5: Queue Management

Analyzes patient categories and waiting times.

Module 6: Doctor Requirement Analysis

Assumption:

1 Doctor = 20 Patients

Calculates doctors required daily and during peak hours.

Module 7: Bed Occupancy Prediction

Assumption:

30% of patients require admission.

Calculates daily and weekly bed requirements.

Module 8: Peak Hour Detection

Identifies:

- Busiest Hour
- Least Busy Hour
- Average Patient Arrivals

Module 9: Resource Forecasting

Predicts:

- Doctors
- Beds
- Emergency Staff

for future patient loads.

9. Algorithms Used

Priority Algorithm

Rules:

- Oxygen < 90 → +30
- Heart Rate > 120 → +20
- Temperature > 102 → +15
- Age > 65 → +10

- Severity Score $\times 5$

Classification Algorithm

Score Range	Category
0–25	Normal
26–50	Moderate
51–75	High Priority
76+	Critical

10. Statistical Analysis

The following statistics are calculated:

Age Statistics

- Mean Age
- Median Age
- Standard Deviation

Health Statistics

- Average Oxygen Level
- Average Heart Rate
- Average Temperature

Hospital Statistics

- Average Waiting Time
 - Maximum Waiting Time
 - Minimum Waiting Time
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11. Visualization

Charts Generated:

- Patient Age Distribution
- Oxygen Level Distribution
- Priority Category Distribution
- Hourly Patient Arrivals
- Waiting Time Analysis
- Bed Occupancy Analysis

Libraries Used:

- Matplotlib
 - Seaborn
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12. Resource Forecasting

The system forecasts:

- Future patient load
- Doctor requirements
- Bed requirements
- Emergency staff requirements

using historical trends.

13. AI Recommendation System

Sample Recommendations:

1. Increase doctors during peak hours.
2. Expand ICU capacity.
3. Allocate additional emergency staff.
4. Reduce waiting times through triage.
5. Improve ambulance coordination.
6. Increase bed availability.

7. Monitor critical patients closely.
 8. Automate emergency alerts.
 9. Implement predictive scheduling.
 10. Optimize patient routing.
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14. Results

The project successfully:

- Generated 1000 patient records.
 - Classified patients by priority.
 - Detected critical cases.
 - Forecasted hospital resources.
 - Generated reports.
 - Created interactive dashboards.
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15. Advantages

- Improved patient management.
 - Reduced waiting times.
 - Better resource planning.
 - Data-driven decision making.
 - Real-time analytics support.
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16. Limitations

- Uses synthetic data.
 - Forecasting is based on assumptions.
 - Does not connect to live hospital databases.
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17. Future Scope

Future enhancements may include:

- Real-time patient monitoring.
 - Machine Learning prediction models.
 - Deep Learning-based risk analysis.
 - IoT sensor integration.
 - Cloud-based deployment.
 - Mobile application support.
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18. Conclusion

The Smart Hospital Emergency Room Analytics System demonstrates how Artificial Intelligence, Machine Learning, Data Science, and Data Analytics can improve hospital operations. The system efficiently manages patient data, prioritizes emergency cases, predicts resource requirements, and provides actionable recommendations for healthcare administrators.

19. References

1. Python Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Streamlit Documentation
5. OpenPyXL Documentation
6. Healthcare Analytics Research Papers